Parametric Multimodal User Memory: Storing What Captions Cannot Carry

Bojie Li
Pine AI

Noah Shi
University of Washington

Abstract

A personalized AI agent needs a *user memory*: a persistent model of who its user is, accumulated across many encounters and consulted on each new one. Today that memory is almost always stored as *text* — transcripts of what was said, captions of what was seen, retrieved by similarity search over those words. This works for the *captionable* slice of a person ("my cat is named Bibi"), but it silently drops the *perceptual* slice that no caption can hold: how a user's voice sounds, how their face reads across lighting and age, how tired they sound today against their own baseline. We measure this loss directly. Across five modalities, a text note written by a strong captioner recovers a perceptual identity at as little as 0.11 of a purpose-built encoder's recall, collapsing toward chance exactly where the discriminating signal is not nameable.

We argue that perceptual memory must be **factored**, not captioned. Remembering a perception is two problems: *what* to remember and where it is (the referent the user means, among others in a scene), and *who* or what it is (an identity that survives changes in lighting, age, room, or recording). We assign each to the component that can solve it. A vision-language model localizes the referent in context; a purpose-built perceptual encoder turns that region into an identity *key*; and the key is stored as a single inline token, read by attention at the model's output head, so recall happens inside generation with no retrieval round-trip. Each component covers the other's blind spot: the VLM localizes but is a weak identity encoder (0.54 on cross-age faces, against 0.81 for a purpose-built face encoder at the same memory size), while the encoder identifies but cannot pick a referent out of a cluttered scene (0.05 whole-scene). Factored, the system recovers oracle-level recognition (agentic localization 0.96, equal to the correct-region oracle), and the recognition core — the encoder's cosine, read in-model — is *training-free*, reproducing the encoder's recall exactly on every frozen model we tried (ten families, 1.5B–14B, tied and untied embeddings, transformer and hybrid-Mamba). On **PerceptMem**, a benchmark of 12 perceptual domains and 1,080 tasks spanning faces, voices, painting style, room acoustics, and tone of voice, we map what such a memory can and cannot hold: perceptual identity is capacity-limited and degrades gracefully (recall $\approx \min(1, k/M)$ per slot), while exact facts are binding-limited and belong in a text store. The perceptual bank sits *alongside* text memory rather than replacing it, and text recall is preserved exactly. An agent with both remembers what its user said *and* what they were like.

Code: https://github.com/19PINE-AI/multimodal-user-memory

1 Introduction

When an AI agent "remembers" something about the person it serves, what does it keep? In almost every system today, *text*. Mem0 [Chhikara et al., 2025], MemoryLLM [Wang et al., 2024], MemGPT / Letta [Packer et al., 2023, Team, 2024], M3-Agent [Long et al., 2025], and the wider conversational-memory line [Wu et al., 2024] turn what the agent saw or heard into transcripts (via ASR [Radford et al., 2022]) and captions (via BLIP-2 [Li et al., 2023] or LLaVA [Liu et al., 2023]), store the fragments in a sentence-encoder vector index [Reimers and Gurevych, 2019], and search them later. Personalized vision-language systems (MyVLM [Alaluf et al., 2024], Yo'LLaVA [Nguyen et al., 2024], MC-LLaVA [An et al., 2024], Online-PVLM [Bai et al., 2025], RAP [Hao et al., 2025]) keep the same textual spine: a *named* concept ("Bibi") is the handle, the perception an afterthought.

This is exactly right for one half of a person. "My cat is named Bibi." "My favourite restaurant is in Paris." "I am vegetarian." These are *captionable*: they pass through text intact and come back whole. But they are not the whole person. Consider what else an attentive companion carries about someone:

- How their voice sounds when they say their own name, and how that differs from a stranger saying it.
- How their face reads today: a little older than last year's photo, but unmistakably them.
- How their brushwork looks: "this is by the painter from Tuesday, even though it's from a different period."
- What their kitchen sounds like: "you're calling from the same room as yesterday."
- How tired they sound today, relative to their baseline last week.

None of this is captionable. "A brown-haired man" does not tell two brown-haired men apart; "the user sounded tired" cannot separate today's tired from last week's. Writing these down throws away the very signal that made them discriminative, and no better caption closes the gap: it is a property of the channel. We confirm this is not rhetorical but measurable (Section 5): a re-identification note written by a strong captioner recovers a perceptual identity at as little as 0.11 of a purpose-built encoder's recall. **Text is a lossy medium for perceptual content, and the loss is fundamental to the modality.**

So a personalized agent's memory has two halves. *Captionable* content (facts, preferences, propositions) belongs in a text store, where sentence-encoder similarity works beautifully. *Perceptual* content — how a face, a voice, a room, or a painting actually *is* — needs a primitive that keeps it in its native modality, and today it has no home. It is either flattened into a caption (and lost) or handed to a recognition tool that returns a bare label like "speaker 47", so the perception never reaches the model. The missing piece is a memory that stores a perception *as a perception*.

Our proposal: factor, don't caption. We introduce a **factored parametric multimodal user memory**. Remembering a perception is two problems, not one. First, *what* to remember and *where* it is: in a real scene the referent the user means (one face among several, the painting being discussed) must be localized in context, which is a joint language-and-vision judgement. Second, *who* or *what* it is: the localized region must be turned into a stable identity that survives changes in lighting, age, room, or recording, which is what a purpose-built perceptual encoder does. We assign each problem to the component that can solve it. A vision-language model localizes the referent (an agentic, context-dependent decision); a dedicated encoder (ArcFace [Deng et al., 2019], ECAPA-TDNN [Desplanques et al., 2020], CLIP [Radford et al., 2021]) turns the localized region into an identity *key*; and the key is stored as a single row in a small per-modality bank bolted onto a frozen language model, paired with the model's own vector for a marker token as the *value*, and read back by attention at the output head. Recall is then a token the model conditions on, produced inside generation with no caption and no retrieval round-trip. Registering an identity is a single tensor append, with no per-user training; recall is constant-time. We call the in-model bank AttMem (for *attention memory*); it borrows the shape of *k*NN-LM [Khandelwal et al., 2020] and Memorizing Transformers [Wu et al., 2022] but aims it at a new target, the persistent perceptual identity of a user, registered in $\mathcal{O}(1)$.

Why factor. Neither component suffices alone, and that is the point. The VLM is a strong localizer but a weak identity encoder: its own vision tokens recognize cross-age faces at 0.54 where a face encoder reaches 0.81 at the same memory size, and a context-conditioned VLM hidden state used directly as the key recovers the referent at only 0.25 (Section 3). The encoder is the opposite: it produces an invariant identity but cannot pick the referent out of a cluttered scene, where encoding the whole image scores 0.05, near chance. Factored, each covers the other's blind spot — the VLM localizes and the encoder identifies — and the system recovers oracle-level recognition (0.96, equal to the correct-region oracle) that neither reaches alone (Figure 1).

A worked example. The user says "remember her" in a photo with three people. A captioning memory writes "a woman, dark hair, mid-30s" and searches; the description fits thousands, so a later match is near chance, and we measure exactly this — a text note recovers a face at 0.20 against the encoder's 0.81, a voice at 0.11 against 0.99 (Section 5). The factored memory instead has the VLM localize the woman the user pointed to, an encoder turn that region into an identity key, and the key enter the bank as one marker token. Months later a new photo of her, older and under different lighting, recalls that token inside the model's forward pass, and "her" is available to whatever the agent is reasoning about. How well the identity matches is exactly the encoder's recall; what the factoring adds is that the right region is found in context, the answer survives the conditions text cannot, and it lives inside the model.

Contributions.

- **Perceptual user memory must be factored.** We argue, and measure, that text-shaped memory cannot carry perceptual user content (a text note recovers a perceptual identity at as little as 0.11 of an encoder's recall), and that the right primitive factors the problem into VLM localization, encoder identity, and an in-model store (Sections 1–3).
- **A factored architecture that recovers oracle recognition in context.** The VLM localizes the referent, a purpose-built encoder identifies it, and the identity is stored as an inline attention token. In cluttered scenes this recovers the correct-region oracle (0.96 on faces, 0.36 on paintings, both equal to oracle) where whole-scene encoding is near chance (0.05), across two visual domains and two encoders (Section 5).

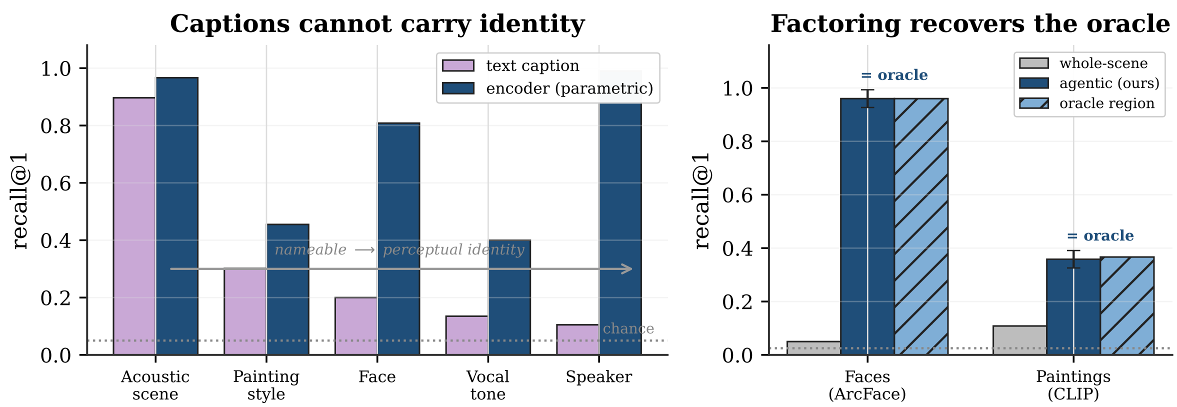


Figure 1. The problem and the factored solution. **Left — captions cannot carry identity.** Across five modalities, a re-identification note written by a strong captioner (purple) recovers a perceptual identity at as little as 0.11 of a purpose-built encoder (navy), collapsing toward chance (0.05, dotted) exactly where the discriminating signal is not nameable. **Right — factoring recovers it.** When the referent sits in a cluttered scene, encoding the whole image is near chance (grey); a vision-language model that localizes the referent before a dedicated encoder identifies it (navy) reaches the correct-region oracle (hatched), on faces (ArcFace) and paintings (CLIP) alike. Both halves are measured in Section 5.

- **A training-free, $\mathcal{O}(1)$ in-model recognition core.** The store is read by attention at the output head with *zero* gradient steps; it reproduces the encoder's recall exactly on ten frozen model families (1.5B–14B, tied and untied embeddings, transformer and hybrid-Mamba), with one read-strength constant as the only knob. Insertion is $\mathcal{O}(1)$ and recall is constant ($\sim$15 ms), 52$\times$ faster than feeding the same memory in as context, which runs out of memory entirely past ten thousand identities (Sections 3–5).
- **PerceptMem and a capacity-law router.** A benchmark of 12 perceptual domains and 1,080 tasks across five modalities (face, painter style, speaker, acoustic scene, tone of voice), with text-only and chance baselines on every comparison, plus two capacity laws that decide what belongs in the perceptual bank versus a text store: perceptual identity is capacity-limited (recall $\approx \min(1, k/M)$), exact facts are binding-limited (Sections 4, 7).

2 Four routes, and the two problems they conflate

Why does the text route fail, not for lack of effort but structurally? Take the voice example in its hardest form: ten samples on file, a new one arriving cross-recording across different rooms and days, and the agent must name which of the ten it is. An agent has four options (Table 1). Three discard the very signal that tells one voice from another. The fourth keeps it — and exposes a second issue every route quietly conflates: recognizing a perception is really *two* problems, localizing the referent in context and encoding its identity, and no single component does both well.

The three that discard the signal. *Caption-and-search* [Chhikara et al., 2025, Wang et al., 2024] writes the voice down ("a male voice, mid-30s, slight Eastern-European accent") and retrieves by similarity over the words. The description is true and useless. It fits thousands of speakers, so cosine over captions cannot separate ten users with even coin-flip reliability. *Recognize-and-label* [Long et al., 2025] calls a recognizer (ArcFace [Deng et al., 2019], ECAPA-

Table 1. Four routes for remembering a perception, plus ours (citations in text). Only routes that keep the raw perception can discriminate across conditions. Embedding retrieval and ours both inherit the encoder's recall exactly; ours additionally localizes the referent with a VLM and reads the identity back *inside* the model as a token the model conditions on, training-free and in $\mathcal{O}(1)$.

Route	What it stores	Keeps percept?	Cross-condition	Register
Caption-and-search	text description	no	chance	$\mathcal{O}(1)$
Recognize-and-label	a bare label	no	label only	$\mathcal{O}(1)$
Per-concept tuning	tuned token/adapter	yes	strong	grad./id
Embedding retrieval	raw embedding	yes	encoder-limited	$\mathcal{O}(1)$
Parametric (ours)	embedding + LM value	yes	= **encoder, in-model**	$\mathcal{O}(1)$

TDNN [Desplanques et al., 2020]) and stores the label it returns ("speaker 47"). The model can recall that a label was assigned, but it never holds the perception and cannot compare today's voice against last week's. *Per-concept tuning* [Nguyen et al., 2024, Alaluf et al., 2024] fine-tunes a token or adapter per concept. It works, but each identity costs a second of gradient descent and yields a concept-specific artifact. That is the wrong economics for a companion registering dozens of characteristics per user, continuously.

The one that keeps it, and what it still needs. *Embedding retrieval* skips the words, indexes the raw voiceprint, and retrieves by cosine over the encoder's vectors. It is the only text-free route that keeps the perception, and the strongest baseline we take seriously throughout. It succeeds exactly to the extent that the encoder is invariant across recording conditions, and fails to the extent that it is not. On a 2180-identity face pool it reaches 0.93 recall at ten identities — strong, but short of certainty, and eroding as the pool grows. Crucially, embedding retrieval presumes the perception has *already* been isolated: someone handed it a clean voiceprint or a tight face crop. In a real scene that isolation is itself a hard, context-dependent step, and it is where a bare encoder is helpless (a single embedding of a three-person photo recognizes the intended person at 0.05, Section 5). This is the second problem the four routes conflate: *localizing* the referent is not the same as *identifying* it, and the strongest text-free route only solves the second.

Our mechanism keeps the encoder's identity step and adds the two things it lacks. It puts a VLM in front to localize the referent the user means, and it changes where the answer lives: it stores the voiceprint as a key but pairs it with the language model's *own* internal vector for a marker token as the value, and reads the pair back by attention inside the model. The recalled value is then a token the model conditions on, not a label handed back from an external index. The recognition is identical to retrieval's — with a sharp read the attention is a hard argmax over the same encoder cosine, so the memory reproduces embedding retrieval exactly, never beating it. The contribution is the factoring around that core: the referent is found in context, the identity survives conditions text cannot, the recall lives inside the model, registration is $\mathcal{O}(1)$, and the whole thing works without modification on any frozen model.

3 The factored architecture

Remembering a perception factors into three steps, and we take each in turn: *localize* the referent in context (the VLM), *identify* it with a dedicated encoder, and *store* the identity as a token read inside a frozen model. The first two are off-the-shelf components placed where they are strong; the third, AttMem, is the mechanism this paper contributes.

Localize: the VLM decides what to remember and where it is. A perception rarely arrives pre-cropped. The user gestures at one person in a group, names the painting on the left, refers to the voice that just spoke; the system must turn that context-dependent reference into a region before anything can be encoded. This is exactly what a vision-language model is good at and an encoder is not: the VLM grounds the referent against the surrounding scene and the user's words, returning a region (a bounding box, a span of audio). We use the VLM's native grounding for this and re-detect inside the returned region where a modality detector exists (RetinaFace for faces, from the same InsightFace project as ArcFace), so box imprecision and un-aligned scenes are handled before identity is read. The necessity of this step is empirical: encoding the whole scene, with no localization, recognizes the intended referent at 0.05 on faces and 0.11 on paintings, near chance, because a single embedding cannot represent a chosen referent among distractors (Section 5).

Identify: a purpose-built encoder, not the model's own tokens. The localized region is read by a dedicated perceptual encoder into an L2-normalized identity *key*: ArcFace for faces, ECAPA-TDNN for speakers, CLIP for painting style, AST [Gong et al., 2021] for acoustic scenes, a wav2vec2 emotion encoder [Baevski et al., 2020] for tone. It is tempting to skip this and read identity from the VLM's own context-conditioned representation — to have the model "output the embedding of the face part" directly. We tested that and it does not hold: a linear probe on a context-conditioned VLM hidden state recovers the referent at 0.25 (raw hidden state 0.14, against an oracle of 0.80), and the VLM's native vision tokens recognize AgeDB cross-age faces at 0.54 where ArcFace reaches 0.81 at the same $N=20$ (Tables 5, 3); the gap persists end-to-end inside a real VLM (§6, Appendix Table 14). The VLM is a strong localizer and a weak identity encoder; the division of labour is not a convenience but a requirement. The key must be a dedicated perceptual encoder, separate from the model's own representation, which is also why the mechanism pairs naturally with an omni model's encoder used as an external feature rather than its in-context tokens.

Store: an in-model attention memory. The identity key is stored and recalled by AttMem, a table of rows on a frozen language model (Figure 2). Row i holds a *key* $\boldsymbol{k}_i$ (the encoder embedding above) and a *value* $\boldsymbol{v}_i$ (the language model's own embedding for a marker token assigned to that identity). Storing the model's native vector as the value, rather than an arbitrary learned one, is deliberate: it makes the eventual logit boost for the right marker clean and self-reinforcing, rather than an accident of two unrelated vectors.

Reading the memory is one step of attention. When a localized perception arrives at generation time, the model forms a query $\boldsymbol{q}$ from its encoder key and, just before the output head, computes

$$\boldsymbol{w} = \mathrm{softmax}\big(\beta\, \boldsymbol{q}^\top \boldsymbol{K}\big), \qquad \boldsymbol{r} = \boldsymbol{w}^\top \boldsymbol{V}, \qquad \boldsymbol{h}' = \boldsymbol{h} + g \cdot W_o\, \boldsymbol{r},$$

where $\boldsymbol{K}, \boldsymbol{V}$ stack the bank's keys and values, β is the attention sharpness, g a residual gain, and W_o the identity. In words: the perception softly looks up the bank, pulls back a blend of the matching markers' model-side vectors, and nudges the next-token prediction toward them. The blend is the entire read, and it adds *no trainable parameters*: β and g are two constants, and the values are the model's own marker embeddings.

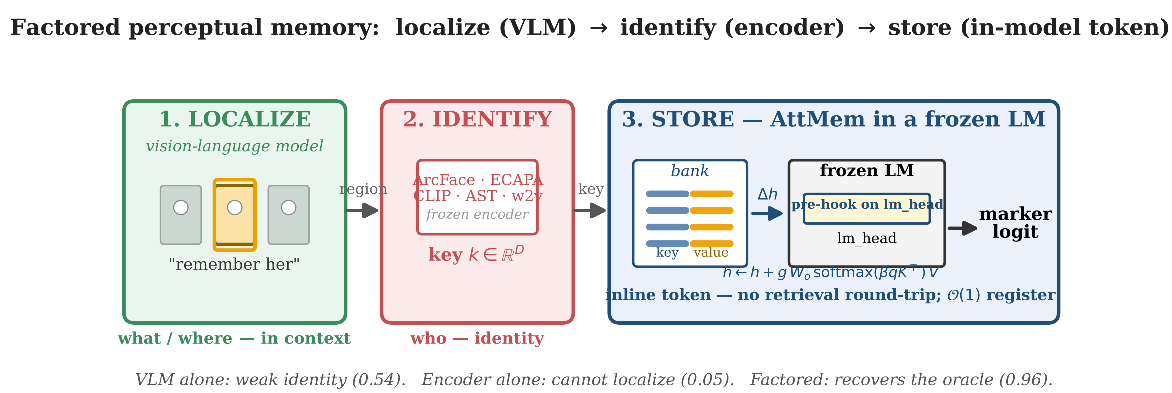


Figure 2. The factored architecture. **(1) Localize:** a vision-language model grounds the referent the user means in a cluttered scene (what / where, context-dependent). **(2) Identify:** a purpose-built frozen encoder (ArcFace, ECAPA, CLIP, ...) turns the localized region into an L2-normalized identity key (who, condition-invariant). **(3) Store:** ATTMEM keeps each key in a per-modality bank on a frozen LM, paired with the model's own marker-token vector as the value, and reads it by a forward pre-hook at the output head that adds a residual biasing the next token toward the matching marker. Registration is a single tensor append; the read has no trainable parameters, only two constants (attention sharpness β, residual gain g). Neither half suffices alone — the VLM is a weak identity encoder (0.54) and the encoder cannot localize (0.05) — but factored they recover the oracle (0.96).

Design choices for a faithful read. Four details determine whether the read recovers the encoder's nearest neighbour cleanly, and getting them right is what lets the mechanism work with no training. (i) *Do not* divide the attention logits by $\sqrt{D}$: with normalized keys, that flattens every cosine difference into near-uniform attention. (ii) Set the sharpness β high, so the attention is a near-hard argmax over the bank rather than a diffuse average. (iii) Attach the hook at the output head, not at an intermediate layer, so the residual reaches the logits undiluted. (iv) Set the gain g large, so the retrieved marker dominates the hidden state and the read is exact; on models whose output embeddings are small in norm, g must be larger still (Section 6). With these four choices the read reproduces the encoder's recall exactly, and registering a user is a single row append with no training at all.

Registration versus fine-tuning. The economics are the point (Figure 3). Adding an identity is a single `torch.cat` (about half a millisecond to add a thousand), and recall is constant-time regardless of bank size, because the lookup is a tiny matrix multiply dwarfed by the model's own forward pass ($\sim$15 ms either way). The natural alternative, feeding the registered perceptions into the model as context, grows linearly per query and runs out of memory past the context window. Our mechanism is 52$\times$ faster at a thousand identities, and is the only one of the two that still functions at ten thousand. Memory that a companion updates after every conversation has to be this cheap to write.

4 The PerceptMem benchmark

To measure perceptual recall we built **PerceptMem**, five sub-modalities each chosen for one reason: the signal that discriminates is provably destroyed by captioning (Table 2). They span

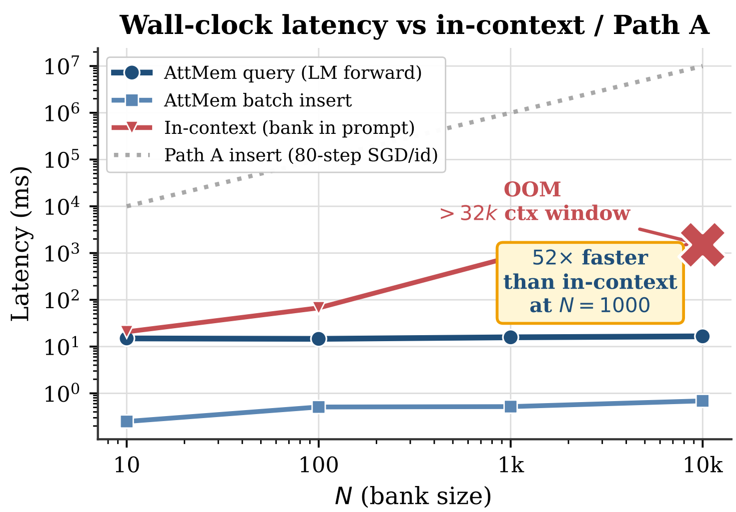


Figure 3. Recall latency is flat in bank size ($\sim$15 ms, dominated by the frozen model's own forward pass) and insertion is $\mathcal{O}(1)$. Feeding the same registered perceptions in as context grows linearly and exhausts the context window past ten thousand identities. Per-concept fine-tuning (Path A-style) costs gradient steps per identity. Log–log axes.

vision and audio, identity and style, the physical and the affective:

Table 2. The five PerceptMem sub-modalities. Each is included because no caption can carry its discriminating signal.

Sub-modality	Source / encoder	Why text cannot carry it
Face across age & lighting	LFW+AgeDB / ArcFace	"brown-haired man" fits thousands
Painter style	WikiArt / CLIP-mid	no caption separates early vs. late Monet
Speaker across recordings	LibriSpeech / ECAPA-TDNN	timbre is not transcribable
Acoustic scene	ESC-50 / AST	"traffic noise" fits every street
Tone of voice	RAVDESS / wav2vec2	"sounded tired" lacks a personal baseline

The sources are all standard: LFW [Huang et al., 2008] and AgeDB [Moschoglou et al., 2017] read with ArcFace [Deng et al., 2019], WikiArt [WikiArt, 2010] with mid-layer CLIP [Radford et al., 2021], LibriSpeech [Panayotov et al., 2015] with ECAPA-TDNN [Desplanques et al., 2020], ESC-50 [Piczak, 2015] with AST [Gong et al., 2021], and RAVDESS [Livingstone and Russo, 2018] with a wav2vec2 emotion encoder [Baevski et al., 2020]. The benchmark deliberately tests *memory*, not perception: every sub-modality has a fixed, off-the-shelf encoder that any method may use as black-box infrastructure, so no one wins by having a better encoder. The interface is just `register` and `recall`. Data is cross-session: a registered sample and its query come from different recordings. The identities used to train the memory never overlap with those it is evaluated on. For a memory of N identities we register one sample each, then issue cross-condition queries and ask the method to name the right one. For tone of voice, each registered "identity" is a (speaker, emotional-state) pair, so recalling it means matching a paralinguistic state across separate utterances, something a per-utterance caption cannot do without the user's own history.

For statistical rigor we expand these five conceptual sub-modalities into a 12-domain, 1,080-task benchmark across multiple datasets and encoders (Section 5, Table 5), and report a text-only baseline and a chance floor on every comparison. Throughout, our reference point is **embedding retrieval**: the same encoder, cosine nearest-neighbour over the registered keys. We call it the *encoder ceiling*, because it is the best one can do with the encoder's own similarity, and our in-model read reproduces it rather than exceeding it.

5 Empirical evaluation

We organize the evaluation around the three claims the factoring makes. First, the half text drops is real and large: captions cannot carry perceptual identity (§5.1). Second, the factored architecture recovers it: VLM localization plus an encoder identity reaches the correct-region oracle in cluttered scenes (§5.2). Third, the recognition core is the encoder, read in-model for free, and that is a feature, not a shortfall (§5.3). We then show the in-model store buys composition with text memory (§5.4) at a cost no external index can match (§5.5), and perturbs nothing (§5.6).

5.1 Captions cannot carry perceptual identity

The baseline a system would otherwise use is a text note: describe the perception in words, store the sentence, and retrieve by text similarity. We test it on *all five modalities* and give it its best shot, paired against the encoder on identical draws. An LLM writes a re-identification note (Qwen2.5-VL for images, Qwen2.5-Omni for audio), a sentence encoder embeds it, and recognition is cosine nearest-neighbour over the notes; the encoder scores the same registrations and queries. The result is a clean router gradient (Table 3, Figure 4). Where the discriminating signal is *nameable*, text nearly matches the encoder: on acoustic scenes a note like "someone typing on a keyboard" recovers 0.897 against the encoder's 0.967. Where it is a perceptual *identity*, text collapses toward chance: speaker 0.105 versus ECAPA 0.990 (a note of "high-pitched voice with an accent" fits countless speakers), vocal tone 0.135 versus 0.400, face 0.200 versus 0.808. Painting style sits between, half nameable, at 0.302 versus 0.455. Text is a competent memory exactly for what can be said and a failing one for what cannot — which is precisely the line the router draws between the text store and the parametric bank (§7).

Table 3. Text-only caption-and-search versus the parametric encoder, paired on identical draws (N=20, 30 draws, 95% CI), across all five modalities. An LLM writes a re-identification note (Qwen2.5-VL for images, Qwen2.5-Omni for audio); a sentence encoder matches it. Rows are ordered by how nameable the signal is. Text nearly matches the encoder on the nameable acoustic scene and collapses toward chance on perceptual identity (speaker, tone, face). This is the router's text/parametric boundary, measured.

Modality (dataset / encoder)	text-only	encoder	chance
Acoustic scene (ESC-50 / AST)	0.897 ± 0.018	0.967 ± 0.010	0.050
Painting style (WikiArt / CLIP)	0.302 ± 0.030	0.455 ± 0.029	0.050
Face (AgeDB / ArcFace)	0.200 ± 0.030	0.808 ± 0.026	0.050
Vocal tone (RAVDESS / wav2vec2)	0.135 ± 0.021	0.400 ± 0.030	0.050
Speaker (VoxCeleb / ECAPA)	0.105 ± 0.016	0.990 ± 0.007	0.050

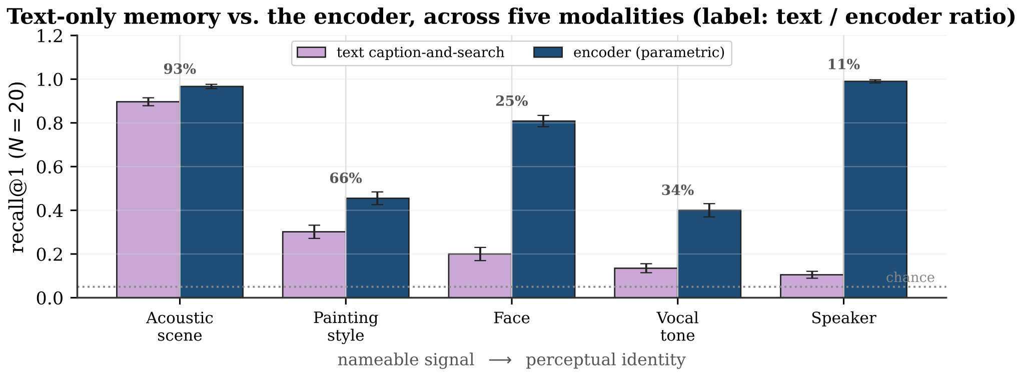


Figure 4. Text-only caption-and-search versus the parametric encoder, paired on identical draws across all five modalities (N=20, 30 draws, 95% CI); the label above each pair is the text/encoder ratio. Ordered by how nameable the signal is. Text nearly matches the encoder on the nameable acoustic scene (93%) and collapses toward chance on perceptual identity (speaker 11%), tracing the router's boundary between the text store and the parametric bank.

5.2 The factored architecture recovers oracle recognition in context

In deployment a perception rarely arrives pre-cropped: the referent sits in a scene among other people or objects, and the system must localize the one the user means before encoding it. Encoding the whole scene fails, because a single embedding cannot represent a chosen referent among distractors (whole-scene recall 0.05 on faces, 0.11 on paintings, near chance). The factored remedy is the architecture of Section 3: the VLM grounds the referent, the region is aligned and read by the perceptual encoder, and the resulting identity key is matched in the bank. On faces (M=40, two per scene, five seeds) the VLM grounds the correct region on every trial (grounding 1.00), and after RetinaFace alignment the ArcFace key recovers the oracle exactly: agentic recall 0.960 ± 0.033 equals the correct-region oracle 0.960 ± 0.033, against whole-scene 0.050 (Table 4, Figure 5). The pipeline is not face-specific. On paintings, with a CLIP encoder and no landmark alignment, the VLM localizes the referenced canvas (grounding 0.98) and agentic recall 0.358 ± 0.033 again equals the oracle 0.367 ± 0.043, against whole-collage 0.108 (three seeds). Localization is lossless and domain-general; what it recovers is bounded by the encoder, modest for CLIP style and near-perfect for ArcFace identity. This is the factoring made visible: the VLM decides *what* to remember and where it is, and a purpose-built encoder decides *who* it is, and neither half alone clears chance on this task.

Table 4. Agentic localization recovers the correct-region oracle across two visual domains and two encoders. The VLM grounds the referent in a multi-object scene; the region is encoded and matched. Whole-scene encoding is near chance; agentic recall equals the oracle within noise. Faces: ArcFace with landmark alignment, 5 seeds. Paintings: CLIP, no alignment, 3 seeds. M=40, two referents per scene; mean±95% CI.

Domain (encoder)	oracle	agentic	whole	grounding	chance
Faces (ArcFace, aligned)	0.960 ± 0.033	0.960 ± 0.033	0.050	1.000	0.025
Paintings (CLIP)	0.367 ± 0.043	0.358 ± 0.033	0.108	0.979	0.025

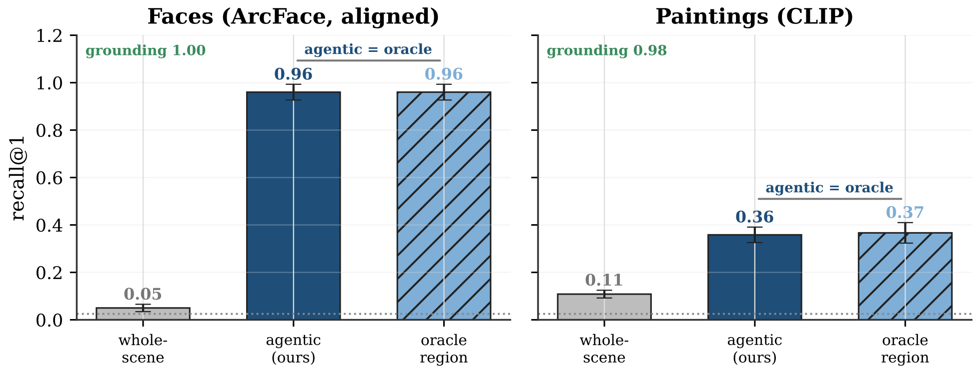


Figure 5. The factored architecture in cluttered scenes, across two visual domains and two encoders (M=40, two referents per scene). Whole-scene encoding (grey) is near chance; the agentic pipeline (navy) recovers the correct-region oracle (hatched) within noise, with the VLM grounding the referent correctly on essentially every trial (grounding annotated). Faces use ArcFace with landmark alignment (5 seeds); paintings use CLIP without alignment (3 seeds).

5.3 The recognition core is the encoder, read in-model for free

Once the referent is localized and encoded, recognition is the bank's job, and the bank reproduces the encoder. With a sharp read the memory's attention is a hard argmax over the same encoder cosine that embedding retrieval uses, so across the five sub-modalities its recall equals the encoder's, neither better nor worse. We measure this with a paired protocol that scores the memory and retrieval on identical registrations and queries across twenty draws and randomises each target's bank slot (Appendix A). Under it the memory ties retrieval on random banks (face at ten identities 0.942 vs 0.948; style at five 0.476 vs 0.473) and, on banks of the nineteen most-confusable look-alikes, stays within a few points (0.773 vs 0.853). A learned metric on the same features (LDA, within-class whitening) lands in the same place (Appendix A): the encoder's cosine is the best ruler these features admit, and the memory inherits it. The value of the in-model read is not a higher number; it is where the number lives.

Breadth and statistical rigor. To test the recognition core at scale we evaluated it over 1,080 tasks: 12 domains spanning the five modalities across multiple datasets and encoders, three working-set sizes $N \in \{10, 20, 40\}$, and 30 independent draws, each carrying a 95% confidence interval (Table 5, Figure 6). Parametric recall clears chance in every domain and *tracks the encoder*: strong purpose-built encoders recognise near-perfectly (ECAPA speaker 0.99, ArcFace/AntelopeV2 face 0.87–0.95), weak ones recognise weakly (CLIP/DINOv2 style 0.24–0.32), and the VLM's own native vision encoder reads the same AgeDB faces at 0.54, far below the purpose-built face encoder on identical inputs — the cleanest statement that the encoder, not the memory, sets each ceiling, and the quantitative core of the "identify" step's design rule (§6). The macro-average is 0.69 against chance 0.05 at N=20 (0.75 and 0.64 at N=10 and 40).

Training-free on any frozen model. Because the read is a similarity-weighted blend of marker tokens, nothing about the recall depends on the host language model, and it ports without training. Across ten families (Qwen2.5 1.5B/7B, Qwen3 4B/8B/14B, Phi-3.5, SmolLM2, DeepSeek-Llama-8B, Mistral-7B, and a hybrid-Mamba Granite) the read reproduces the en-

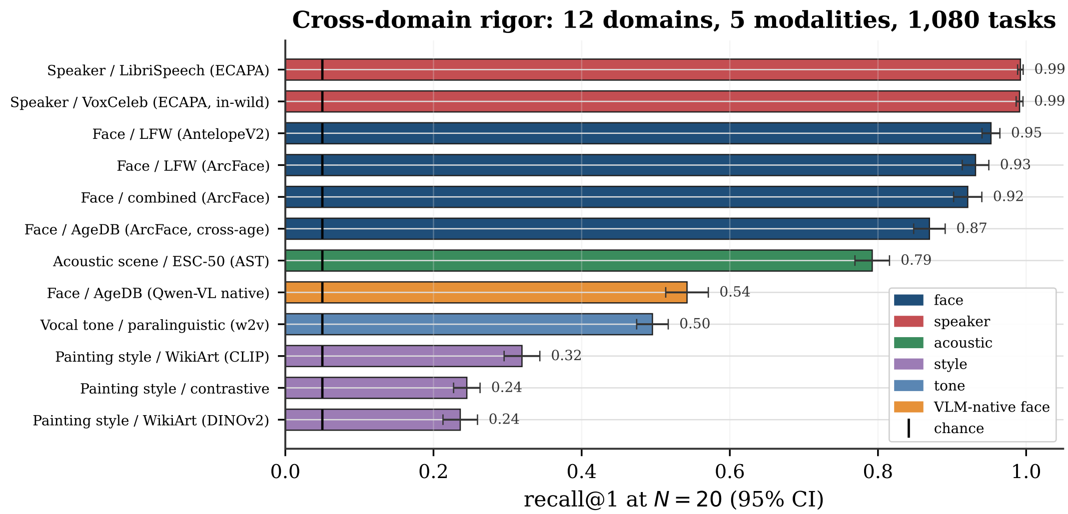


Figure 6. The recognition core across 12 perceptual domains and 5 modalities (1,080 tasks; recall@1 at N=20, 30 draws, 95% CI), coloured by modality, against chance ($1/N$, black ticks). Parametric recall tracks the encoder — near-perfect for strong encoders (speaker, face), lower for weak ones (style, tone) — and clears chance in every domain. The VLM's own native vision encoder (orange) reads the same AgeDB faces far below the purpose-built face encoder, the cleanest statement that the encoder sets each ceiling.

Table 5. Cross-domain recognition recall@1 at N=20 registered identities (30 draws, 95% CI), parametric memory (encoder cosine = the attention read) versus chance, across 12 perceptual domains and 5 modalities. Recall tracks the encoder; it clears chance everywhere. The full benchmark is 1,080 tasks ($N \in \{10, 20, 40\}$). The starred row is the VLM's own native vision encoder on the same AgeDB faces as the first row, far below the purpose-built face encoder.

Domain	Modality	#id	param	chance
Face / AgeDB (ArcFace, cross-age)	face	500	0.869 ± 0.021	0.050
Face / LFW (ArcFace)	face	1680	0.932 ± 0.018	0.050
Face / LFW (AntelopeV2)	face	901	0.953 ± 0.012	0.050
Face / combined (ArcFace)	face	2180	0.921 ± 0.019	0.050
Speaker / LibriSpeech (ECAPA)	speaker	58	0.992 ± 0.004	0.050
Speaker / VoxCeleb (ECAPA, in-wild)	speaker	40	0.991 ± 0.005	0.050
Acoustic scene / ESC-50 (AST)	acoustic	50	0.792 ± 0.023	0.050
Painting style / WikiArt (CLIP)	style	128	0.319 ± 0.024	0.050
Painting style / WikiArt (DINOv2)	style	50	0.236 ± 0.023	0.050
Vocal tone / paralinguistic (w2v)	tone	168	0.496 ± 0.021	0.050
Painting style / contrastive	style	26	0.245 ± 0.018	0.050
Face / AgeDB (Qwen-VL native)*	face	567	0.542 ± 0.029	0.050

coder's recall (Appendix Figure 9), and a five-architecture by five-modality grid holds to within one point in every cell (Appendix Table 9). On tied-embedding models the match is *exact on every draw*; untied models reach exact recall with one larger read-strength constant; the hybrid-Mamba reader is within a point. This universality is a supporting property of the store, not the headline: it says the "store" step is free to host on whatever model the agent already runs.

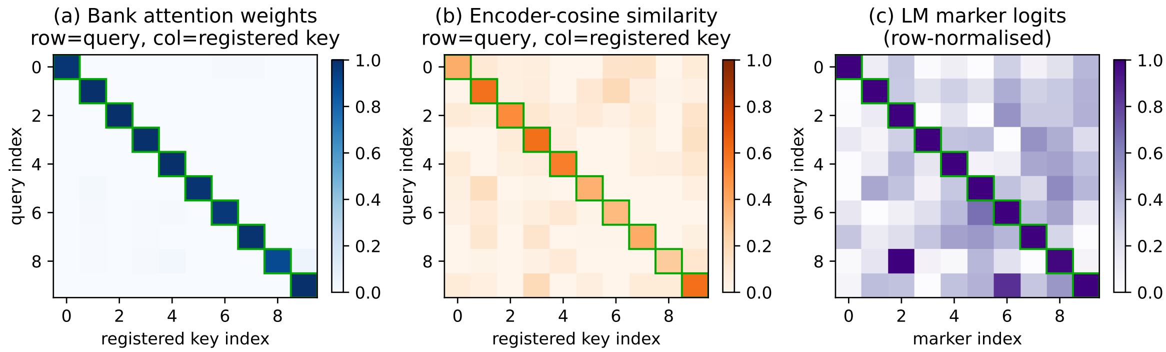


Figure 7. One recall, three views (held-out faces, ten identities). **(a)** with a sharp temperature, bank attention is a near-pure diagonal (avg. 0.98), committing to the top encoder match. **(b)** the underlying encoder cosine has a softer diagonal (avg. 0.46) with off-diagonal mass; the memory inherits whatever errors live here. **(c)** the resulting marker logits keep the argmax identity on top. The attention faithfully reads the encoder's nearest neighbour as a token; it does not sharpen the similarity.

5.4 A recall the model can think with

The reason to recall a perception as a token, rather than a label from an index, is that the token participates in the model's reasoning. We test this directly. We register M faces, each with the person's real first name as its marker, and write one fact per name into the prompt context ("Anna is a teacher. Mark is a doctor."). Then we show a held-out, cross-condition photo and ask for the fact. The whole chain runs inside one model in a single pass: the face recalls the name through the bank, and the name retrieves the fact from context. At ten identities the end-to-end accuracy is 0.80, against a chance baseline of 0.10 when the face is withheld (Table 6). The recognition leg equals the encoder's recall (0.91); what bounds the chain as M grows is the host model's own in-context lookup, not the memory. Here the perception is already a token the model conditions on, folded into the forward pass the agent is running anyway — something an external retrieval index, which returns a label outside the model, does not provide. Figure 7 opens up a single read: a sharp temperature concentrates the encoder's speckled cosine matrix into a near-one-hot diagonal, so the marker logits put the argmax on the same identity retrieval would pick, recovered as a decisive token rather than a diffuse blend.

Table 6. In-model composition: register M faces with name markers, bind one fact per name in context, then recall the fact from a held-out face. "recog" is face→name (the bank, = encoder recall); "compose" is the end-to-end face→name→fact chain; "blind" withholds the face (chance $1/M$). 12 draws, mean±std. Composition tracks recall times the model's in-context lookup, and beats blind chance by 4–10×.

M	recog (memory)	lookup (text)	compose (e2e)	blind (chance)
5	0.95 ± 0.09	0.82	**0.77** $\pm$ 0.19	0.20
10	0.91 ± 0.10	0.88	**0.80** $\pm$ 0.06	0.10
15	0.87 ± 0.08	0.68	**0.59** $\pm$ 0.18	0.067
20	0.86 ± 0.11	0.57	**0.51** $\pm$ 0.14	0.05

5.5 Cost: ties an index on accuracy, wins on architecture

The natural baseline is retrieve-and-reprompt: cosine-retrieve the nearest registered face, recover its name, and re-prompt the model. Because both routes recognise with the same encoder cosine and reason with the same frozen model, they reach *identical accuracy*, and a careful comparison confirms it: composition is 0.81 either way, and a multi-perception "how many distinct people are here?" task is 1.00/1.00 at two or three faces and 0.70/0.70 at four (Table 7). On cost the pipeline is in fact *cheaper* for pure recognition, since counting distinct identities needs only cosine and no model forward at all. Against this strong baseline the parametric memory is not a faster or more accurate recogniser. Its advantage is architectural: recognition folds into the generation the agent is already running, with no external index, no second prompt, and $\mathcal{O}(1)$ registration, so one frozen model is the whole system.

The comparison sharpens against the baseline a VLM actually reaches for: keeping the perception in context as image tokens. A face crop costs 64 vision tokens in Qwen2.5-VL at 224px and 256–576 at higher resolutions (a full image approaches a thousand), where AttMem reads the same perception as *one* marker token. That is a 64–576$\times$ reduction in prefill tokens per perception, and because the bank lives outside the context the memory scales without bound while image-in-context exhausts a 128k window at about two thousand crops. On accuracy this baseline is also far weaker, since the VLM recognises with its general vision tokens (0.60 end-to-end, §6) rather than a face encoder (0.93). So the parametric memory ties the external-index pipeline and outperforms the in-context-image baseline on both accuracy and cost. That, together with the capacity-law router, is the contribution, not a benchmark number.

Table 7. In-model memory vs. a retrieve-and-reprompt pipeline (Qwen2.5-3B, 10 draws). Accuracy is identical everywhere, because both recognise with the same encoder cosine. The pipeline is cheaper for pure recognition (the distinct-count task needs no model forward); the in-model read folds recognition into the model's forward instead of an external index.

Task	In-model acc	Pipeline acc	In-model lat.	Pipeline lat.
Compose (face$\to$fact)	0.81	0.81	122 ms	98 ms
Count distinct, K=2	1.00	1.00	238 ms	0.09 ms
Count distinct, K=3	1.00	1.00	361 ms	0.09 ms
Count distinct, K=4	0.70	0.70	472 ms	0.09 ms

5.6 Non-regression on text

The perceptual memory is meant to sit beside an ordinary text memory, not perturb it. With the bank installed and populated, the model's next-token predictions on text-only prompts are *byte-for-byte identical* to the untouched model (top-1 match on every probe), because the hook adds a residual only at perceptual positions. Register faces and voices together and the two banks stay independent without any training (cross-modal leakage 0.017 and 0.067). An agent can keep "my favourite restaurant is in Paris" in its text store and "how Bibi looks across lighting and age" in its perceptual bank, and lose nothing on either. The discrete-codebook design we tried before this one, which snapped each perception to one of K learned codes, never escaped $\sim$0.07 recall past a few hundred identities for the same reason captioning fails — a categorical bottleneck discards the signal the encoder preserved; we keep it as a contrast in

Appendix C.

6 The encoder is the lever

The factoring puts identity on a dedicated encoder, and that choice dominates everything downstream: because the read reproduces the encoder's recall, the memory's quality is the encoder's quality, and the engineering reduces to a few rules.

Choose the encoder well; nothing in-model recovers what it discards. The memory inherits the encoder's cross-condition invariance and nothing else, so recall moves with encoder quality on the *same* identities. On AgeDB, ArcFace reaches 0.95 recall at ten identities and 0.75 at a hundred, while a general vision-language model's own vision tokens manage only 0.64 and 0.34 on those exact faces; on LFW, ArcFace and AntelopeV2 agree to within two points. A learned metric on the same features does no better (Appendix A), and at large pools the encoder's own cosine is the best anyone does. If a task needs better recall, the lever is a better encoder.

Use an external encoder for the key, not the model's native tokens. We run the memory end-to-end inside a real vision-language model (Qwen2.5-VL) that sees raw AgeDB face crops through its own visual front-end. The mechanism works — register one crop, recognise a cross-condition crop, all inside the VLM — but the recall is only the VLM's native-token recall, weak because those tokens are not face-specialised: 0.60 at ten identities, matching cosine over the same native keys (0.50) and far below a purpose-built encoder (Appendix Table 14). This is why the "identify" step uses a dedicated encoder and the VLM is reserved for the "localize" step it is actually good at: **the key should be a dedicated perceptual encoder, separate from the model's own representation**, which is also why the mechanism pairs naturally with an omni model's encoder used as an external feature rather than its in-context tokens.

The store is universal across frozen models, with one constant. The read ports to any frozen model without training (Appendix Figure 9, Table 9); the only model-dependent choice is the read-strength gain. On tied-embedding models the default suffices and the match to the encoder is exact on every draw; on untied models the marker rows are smaller in norm, so the gain is raised once, a constant set by inspection. The constant is forgiving rather than narrow: tied models are exact across the 64× range from 16 to 1024, untied models above a threshold (Mistral exact and stable from 256 through 1024, Appendix Table 10), and a non-transformer reader (hybrid-Mamba Granite) holds within a point. No family failed, and because nothing is trained, there is no convergence to be fragile.

7 Discussion

The claim is structural, and the numbers illustrate it. The argument is not "our method scores higher." It is that text is the wrong container for half of what an agent should remember about a person, that recognizing a perception factors into localizing it and identifying it, and that a frozen model fronted by a VLM and backed by an encoder bank solves both halves where text and a bare encoder each solve neither. The empirical results matter because they pin each piece: the text loss is real and modality-graded (Table 3), the localization step is necessary and the factored system recovers the oracle (Table 4), and the recognition core is the encoder's recall given an in-model home, free and universal (Appendix Figure 9). The contribution is the factored container, not a recall score; the score is the encoder's.

This is additive infrastructure, not a replacement. The perceptual memory is designed to bolt onto the text memory that agents already have [Chhikara et al., 2025, Wang et al., 2024, Packer et al., 2023, Team, 2024], and the non-regression result is what makes that safe: text recall is preserved exactly, and the perceptual bank activates only at perceptual moments. As long-term memory benchmarks like LongMemEval [Wu et al., 2024] grow to include perceptual content, the gap this paper argues for becomes directly measurable.

The captionable half is going parametric too. This paper is one half of a larger bet: a personalized agent should keep its memory *inside* the model, not in an external index. A companion line makes the same bet for the *captionable* half. *User as Engram* [Li, 2026c] writes per-user *facts* into a hash-keyed parametric memory, showing that the obvious alternative (a per-user LoRA adapter) is *reasoning-negative*: it memorizes facts but degrades the reasoning that uses them, because a LoRA edit is global where a row insertion is local. *User as Code* [Li, 2026b] pushes the same half into executable typed state, and *Programmable KV Cache* [Li, 2026a] keeps it as editable, composable notes in the attention cache rather than in weights or an external index. Our perceptual counterpart shares that thesis with a different mechanism, continuous cross-attention over encoder banks rather than hash-keyed rows. That continuous channel, carrying a perception as a vector projected into the model's own embedding space rather than as text, is the same bet *The Latent Bridge* [Li and Shi, 2026] makes for inter-model communication, where a learned latent link beats writing and re-reading text. The two halves compose into one memory that holds both what the user said and what they are like.

The router: what a latent memory can and cannot hold. The split between the two stores is not a convention. It follows from two capacity laws that we measured by compressing content into k soft tokens read by a frozen language model (Figure 8). Perceptual identity is capacity-limited and degrades gracefully. Compressing M registered identities into k prototype slots and recognising a cross-condition query keeps a k/M fraction of them distinguishable, on faces, voices, acoustic scenes, painting styles, and tones of voice alike (Appendix Table 11). One slot per identity recognises as many as the encoder can, and fewer slots lose identities in proportion. The general law is recall $\approx \min(1, k/M) \cdot C(M)$, where $C(M)$ is the encoder's own recall at M registrations: near 1 for strong encoders (face, voice), lower for the weaker style and tone encoders, so the slot-compression factor $\min(1, k/M)$ is what is universal and the ceiling is the encoder's. This is not an artifact of k-means: a compressor whose k slots we instead learn by gradient descent lands on the same curve, because k hard slots can keep at most k of M identities separable (a pigeonhole bound). Exact factual content behaves in the opposite way. A latent holds one exact code and reads it back (0.87 for a six-character code), but storing several and retrieving one by name collapses at once (0.06 at two codes, near zero beyond), and adding latent tokens does not help. Recognition is a capacity problem, where a latent buys one identity per slot; exact recall is a binding problem, where a latent cannot match a key to its value at any budget. That is why the captionable half wants a text store, which performs that lookup exactly, and the perceptual half wants a parametric bank, which scales with its slots — and it is the same line the text ablation drew empirically (Table 3), now derived.

Limitations. (i) The memory cannot exceed the encoder. Its recall *is* the encoder's, so at very large memories it declines exactly as the encoder's cosine does (0.77 at a thousand faces), and no in-model trick recovers more. (ii) We have measured real cross-condition data to roughly

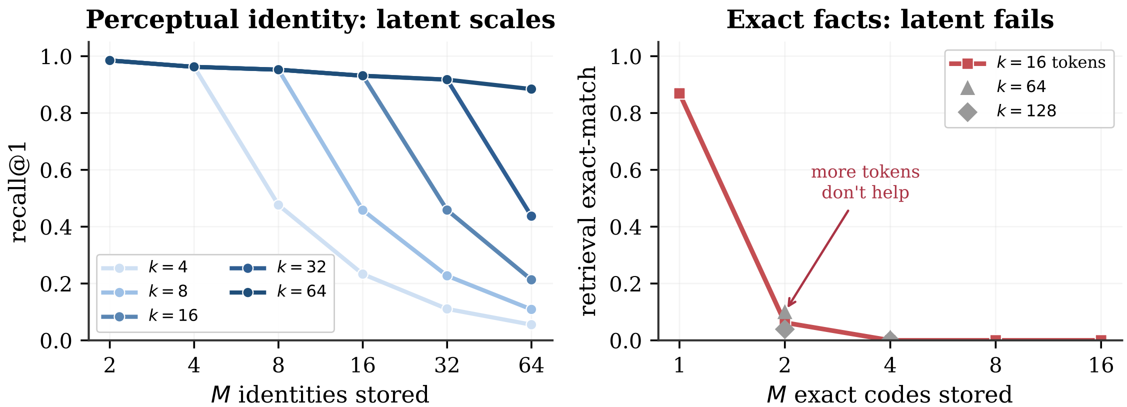


Figure 8. Two capacity laws for memory compressed into k soft tokens read by a frozen language model, the mechanism behind the router. **Left:** perceptual identity is capacity-limited. Compressing M faces into k prototype slots and recognising a cross-condition query gives recall $\approx \min(1, k/M)$, so a slot per identity recognises everyone and more slots hold more people (voices behave identically). **Right:** exact factual content is binding-limited. A latent holds one exact code (0.87) but multi-code retrieval collapses at two (0.06) and adding latent tokens does not rescue it. Recognition scales with the latent budget. Exact recall does not, so it belongs in a text store.

two thousand identities; the constant-time behaviour is confirmed to ten thousand, but recall at that scale awaits data. (iii) The localization step inherits the VLM's grounding: we measure it lossless on two-to-three-referent scenes ($K \leq 4$), and denser or more ambiguous scenes will stress it. (iv) Inside a VLM, native vision tokens fail as keys and an external, modality-specific encoder is required. (v) The one per-model constant (read-strength gain) is set by inspection; untied-embedding models need a larger value, which we have not automated. (vi) The closest prior systems (MyVLM [Alaluf et al., 2024], Online-PVLM [Bai et al., 2025]) have released neither code nor checkpoints; we compare against charitable reimplementations of their core mechanisms (Appendix A, Table 13).

8 Conclusion

A photograph and a written description of a face are not two encodings of the same thing. The description has already thrown away what lets you recognize the person. Agent memory today keeps only the description; we have argued it must keep the photograph too, and that keeping it well means factoring the job. A vision-language model localizes the referent the user means; a purpose-built encoder turns it into an identity that survives the conditions text cannot; and a frozen model stores that identity as a content-addressable row it can read inside its own forward pass. Each component does what the others cannot — the VLM cannot identify, the encoder cannot localize, and an external index cannot hand the answer back inside the model — and together they recover oracle-level recognition where every single piece is near chance. The recognition itself is the encoder's, given a home: training-free, constant-time, and portable to any frozen model. The two memories are complementary, text for what the user *said* and parametric multimodal memory for what the user *is like*, and an agent with both strictly dominates one with only text. That is where a face stops being a sentence, and starts being a memory.

Acknowledgements

We thank Xiangyu Zhang from StepFun AI for the initial ideas that inspired this paper; he introduced the problem of multimodal user memory to the authors and pointed out the limitations of purely textual memory. This paper was produced using Pine Copilot's voice-directed *whisper coding* workflow [Pine AI, 2026], in which the authors specify, discuss, and review the work by voice while a coding agent (Claude Code with Claude Opus 4.8) carries out the planning, coding, experiments, and paper writing. We thank BSQL Networking for hosting the NVIDIA RTX PRO 6000 GPU.

A Detailed results

This appendix collects the per-cell numbers behind Sections 5–6. Recall is top-1 with the argmax restricted to registered markers.

Paired evaluation protocol. A comparison between the memory and retrieval requires two controls: both methods score on *identical* registrations and queries, and each target identity occupies a randomly chosen bank slot. The second control matters because a fixed target slot lets the constant marker token in that slot accrue a baseline output-logit, which inflates recall independently of the query. We score both methods on the same twenty draws with the target slot randomised, and report mean±std. Table 8 gives the result. The memory ties retrieval on random banks, since with a sharp read it computes the same argmax over encoder cosine, and on banks of the nineteen most-confusable look-alikes it trails by a few points: its similarity-weighted blend of marker values carries no per-identity signal beyond the cosine, so it cannot separate confusable identities the encoder already conflates.

Table 8. Paired evaluation: the memory and retrieval scored on identical registrations and queries across 20 draws, with the target slot randomised. The memory ties retrieval on random banks and trails it by a few points on look-alike banks. Recall is the encoder's throughout.

Regime	Cell	Retrieval	AttMem (mean±std)	Δ
random	Face, N=10	0.948	0.942 ± 0.035	-0.6pp
random	Style, N=5	0.473	0.476 ± 0.116	$+0.2$pp
random	Style, N=10	0.428	0.357 ± 0.081	-7.2pp
adversarial	Face, K=19	0.853	0.773 ± 0.008	-8.0pp

Universality grid. Table 9 runs the training-free read across five model architectures and all five modalities, scoring each cell against the encoder with the same paired protocol (20 draws). Every one of the twenty-five cells reproduces the encoder's recall to within one point (worst case 0.010, most cells ≤ 0.003). Tied-embedding models use the default read-strength gain; untied and hybrid models use a single larger value (256), set once per model and held fixed across modalities. The read therefore inherits the encoder's recall regardless of the host model's family, embedding scheme, scale, or even whether it is a transformer.

Read-strength robustness. The one per-model constant, the residual gain, is set by inspection rather than search because exact recall holds across a wide range (Table 10, face N=50, paired).

Table 9. Universality grid: maximum $|\Delta|$ between the training-free memory and the encoder over $N \in \{5, 10, 20\}$ (paired, 20 draws), for five architectures × five modalities. Every cell is within one point of the encoder. "gain" is the single per-model read-strength constant. Encoder recall@1 at N=10: Face 0.95, Speaker 0.99, Acoustic 0.86, Tone 0.52, Style 0.43.

Model	Emb.	gain	Face	Speaker	Acoustic	Tone	Style
Qwen2.5-7B	tied	64	0.001	0.000	0.002	0.003	0.001
Phi-3.5-mini	tied	64	0.001	0.000	0.000	0.002	0.002
DeepSeek-Llama-8B	untied	256	0.000	0.000	0.002	0.002	0.000
Mistral-7B	untied	256	0.000	0.000	0.003	0.010	0.000
Granite-4.0 (Mamba)	hybrid	256	0.003	0.000	0.003	0.001	0.000

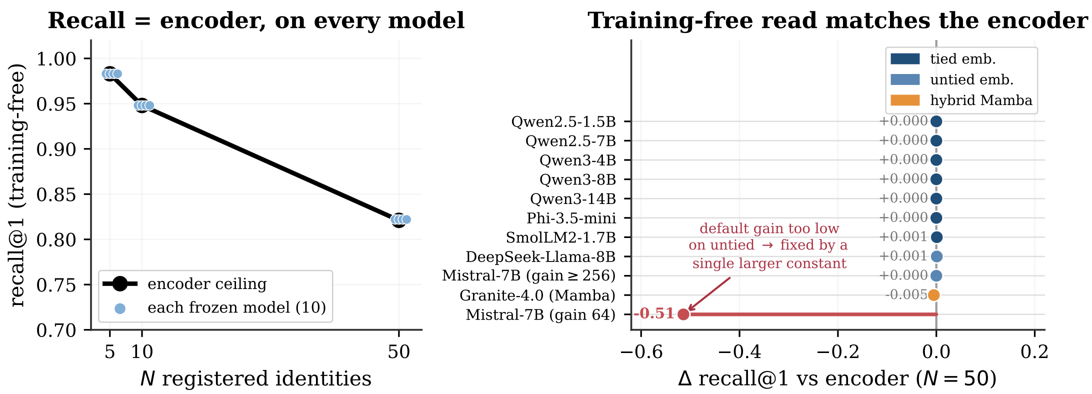


Figure 9. The training-free read reproduces the encoder on every frozen model, the supporting property behind the "store" step. **Left:** each model's training-free recall lands on the encoder ceiling (paired, 20 draws). **Right:** the gap from the encoder at N=50 is ≈ 0 across ten families (1.5B–14B, tied and untied embeddings, hybrid-Mamba); only an untied model at the default gain dips (red), fixed by a single larger read-strength constant. No gradient steps anywhere.

A tied-embedding model is exact at every gain from 16 to 1024; an untied one (Mistral) is below the encoder until a threshold near 128–256 and then exact and stable through 1024. Any value in the upper range works for either, so there is no tuning to get wrong.

Table 10. Read-strength robustness: training-free recall@1 (face, N=50, encoder ceiling 0.821) as the residual gain sweeps 16–1024. Tied models are exact throughout; untied models are exact above a threshold and stay exact. The constant tolerates a wide range.

Model	16	32	64	128	256	512	1024
Qwen2.5-3B (tied)	0.821	0.821	0.821	0.821	0.821	0.821	0.821
Mistral-7B (untied)	0.022	0.060	0.307	0.768	0.821	0.821	0.821

Capacity law across modalities. Table 11 runs the perceptual capacity probe (compress M registered identities into k prototype slots, recognise a cross-condition query, 5 seeds) on all five modalities. The slot-compression factor $\min(1, k/M)$ fits every modality once normalised by the one-slot-per-identity ceiling $C(M)$: the normalised error is ≤ 0.012 on the strong-encoder modalities and ≤ 0.13 on the weaker style and tone encoders. The ceiling $C(M)$ itself is the encoder's recall and varies accordingly, from near 1 on voice to 0.26–0.36 at $M{=}32$ on style and tone. Recognition is capacity-limited and graceful in every modality; only the ceiling moves.

Table 11. Perceptual capacity law per modality. $C(M)$ is the one-slot-per-identity ceiling (recall at $k{=}M$), i.e. the encoder's recall at M. "fit" is the mean $|\,\text{recall}/C(M) - \min(1, k/M)\,|$ over the $k{<}M$ cells: the slot-compression factor is universal; the ceiling is the encoder's.

Modality	$C(2)$	$C(8)$	$C(32)$	**fit to** $\min(1, k/M)$
Face (ArcFace)	0.98	0.95	0.92	0.002
Voice (ECAPA)	1.00	1.00	0.99	0.001
Acoustic (AST)	0.95	0.87	0.76	0.012
Style (CLIP)	0.76	0.46	0.26	0.070
Tone (wav2vec)	0.94	0.67	0.36	0.125

Learned-metric baseline. A natural question is whether *any* re-encoding of the same features beats raw cosine, even if our read does not. We fit a similarity on the same encoder features and the same identities (the 50/50 split, seed 42) and score it with the identical protocol. Two closed-form metrics suffice: regularised LDA and within-class whitening. Table 12 shows they land on raw cosine at small memory and *below* it past a few hundred identities. So there is no discrimination left to extract from these features: the encoder's cosine is already the best ruler, which is why the memory's inheriting it exactly is the ceiling, not a shortfall. A contrastive linear head we also trained underperformed both closed-form metrics and is omitted as a weak instantiation.

Table 12. Learned-metric baseline, recall@1, same encoder and same split as AttMem. A learned metric (LDA on faces, whitening on style) matches AttMem's small-N lift over raw cosine; at large N raw cosine wins for all. Style AttMem here is the seed-42 single run (the multi-seed mean is 0.64 at $N{=}5$); the comparison is within seed.

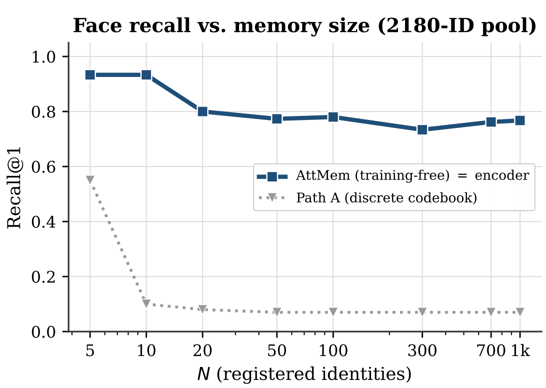
Modality	N	Raw cosine	LDA	Whitening	AttMem
Face (ArcFace)	5	0.933	**1.000**	0.933	0.933
Face (ArcFace)	10	0.933	0.967	0.933	**1.000**
Face (ArcFace)	300	0.734	0.667	**0.744**	0.641
Face (ArcFace)	1000	0.767	0.724	**0.776**	—
Style (CLIP-mid)	5	0.400	0.200	**0.600**	0.467
Style (CLIP-mid)	10	0.400	0.333	0.367	**0.467**

Comparison with per-concept methods. The personalized-VLM line (MyVLM, Yo'LLaVA, Online-PVLM) registers a concept by *learning* something per concept (a classifier head, a soft token, a projection). Their code and checkpoints are unreleased, so we reimplement each method's core mechanism charitably on the cross-condition face task (ArcFace keys, same split). Table 13 reports accuracy and registration cost. Two findings. First, a per-concept linear classifier (MyVLM's mechanism) is statistically indistinguishable from raw cosine at every memory size, because with one registration embedding per identity the trained head collapses to the embedding itself. It buys no accuracy over retrieval, and it costs per-identity gradient descent that grows to 4.5 s at a thousand identities, against our $\mathcal{O}(1)$ tensor append. Second, a learned projection trained with supervised contrastive loss (Online-PVLM's mechanism) *underperforms* raw cosine by 15 to 28 points on this task. We read this as a caution that a learned re-encoding is not free: tuned on identity-disjoint data, it can distort the very cross-condition geometry it was meant to sharpen. No method here, ours included, beats raw cosine on accuracy at scale; the parametric memory's case rests on cost and in-model behaviour, not recall.

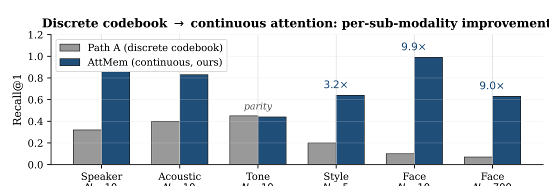
Table 13. Charitable reimplementations of per-concept methods on cross-condition face recall (ArcFace, recall@1, single eval draw). MyVLM's per-concept classifier ties raw cosine but its registration cost grows with the memory; Online-PVLM's learned projection underperforms raw cosine. AttMem leads only at small N and trails at scale (the encoder-ceiling inversion).

N	Raw cosine	MyVLM	Online-PVLM	AttMem	MyVLM reg. cost
5	0.933	0.933	0.800	0.933	0.009 s
10	1.000	0.967	0.750	0.992	0.019 s
50	0.767	0.753	0.675	0.733	0.178 s
100	0.780	0.770	—	0.742	0.528 s
300	0.728	0.725	—	0.637	1.39 s
1000	0.776	0.778	—	0.594	4.52 s

Scaling and the codebook comparison. Figure 10a traces recall against memory size on the face pool. The training-free memory tracks the encoder's own curve (it *is* that curve), declining gracefully from near-ceiling at ten identities to 0.77 at a thousand, while the discrete-codebook predecessor (Path A) flatlines at $\sim$0.07 regardless of codebook size or budget. Figure 10b makes the same point across all five sub-modalities: replacing a discrete codebook with continuous attention lifts recall 2–10$\times$, because a categorical bottleneck discards the signal the encoder preserved. This is the one place a design choice in the read matters; everything else inherits the encoder.



(a) scaling on the face pool



(b) codebook vs. continuous attention

Figure 10. Left: the continuous memory degrades gracefully with memory size (tracking the encoder ceiling), while the codebook predecessor never gets off the floor. Right: continuous attention beats the discrete codebook 2–10$\times$ on every sub-modality.

Vision-language model, key/value orthogonality. Table 14 gives the numbers behind the structural rule of Section 6. Each memory row is compared to retrieval *in its own key space*. An external ArcFace key (orthogonal to the model's hidden space) reproduces the win. The VLM's native vision tokens (already in hidden space) only tie native-token retrieval and stay far below the ArcFace ceiling throughout, so they are the wrong key regardless of method — the empirical basis for reserving the VLM for localization and an external encoder for identity.

Table 14. Parametric memory inside Qwen2.5-VL-3B: same frozen model, only the key encoder changes. Recall at three memory sizes.

Configuration	N=10	N=100	N=1000
Retrieval over ArcFace embeddings (encoder ceiling)	0.933	0.780	0.767
AttMem + external ArcFace keys (orthogonal to hidden space)	**1.000**	0.710	0.494
Retrieval over native vision tokens	0.50	0.35	—
AttMem + native vision tokens (co-located with hidden space)	0.60	0.11	—

B Anatomy of a single recall

It helps to watch one recall end to end. Suppose the agent has met ten people and wants to register an eleventh from a single photo. Registration is one row appended to the face bank (Listing 1). The encoder turns the photo into a 512-dimensional ArcFace embedding, the *key*. The model's own embedding for a freshly assigned marker token becomes the *value*, a vector of the model's hidden width (2048 for Qwen2.5-3B). There is no gradient step and no fine-tuning. The row is the memory.

```
# Register identity #11 from a single photo -- no training:
k = l2_normalize(arcface(photo))        # key:   R^512  (encoder space)
m = assign_marker_token()               # e.g. "<id_11>"
v = model.input_embedding[m]            # value: R^2048 (model's own space)
bank.K = torch.cat([bank.K, k[None]])   # O(1) append
bank.V = torch.cat([bank.V, v[None]])
```

Listing 1. One registered identity is one row. The key is the encoder's view of the perception; the value is the model's own vector for the marker token that names it. Adding the row is a single tensor append.

At recall, a new photo of one of the eleven arrives under different lighting and a year older. Its ArcFace embedding becomes the query $\boldsymbol{q}$. Just before its output head, the model scores $\boldsymbol{q}$ against all eleven keys, softmaxes into attention weights $\boldsymbol{w}$, and pulls back the weighted blend of values $\boldsymbol{r} = \boldsymbol{w}^\top \boldsymbol{V}$. Because each value is the model's own vector for a marker token, adding $g\, W_o \boldsymbol{r}$ to the hidden state lands as a clean boost on the right marker's logit. The model's next token *is* the remembered identity. The whole step is a few matrix multiplies dwarfed by the model's forward pass.

The 0.98-vs-0.46 diagonal gap of Section 5 (Figure 7) is measured on exactly this probe: a held-out ten-identity bank, comparing the sharp attention weights against the raw encoder cosine they are computed from. The attention concentrates the cosine onto its top match, which is why the read recovers the encoder's nearest neighbour as a single decisive marker rather than a diffuse blend.

C The discrete-codebook predecessor

Before continuous attention we spent sixteen development cycles on PATH A: a discrete codebook that snaps each perception to one of K learned codes and remembers the code. It is the natural "compress the perception into a symbol" design, and it is a learned cousin of captioning, which is why its failure is instructive rather than embarrassing.

No setting broke it. Across codebook sizes $K \in \{128, 256, 512, 1024\}$, even after *100,000* steps of continual pretraining on the expanded face pool, top-1 recall at 300 identities stays pinned between 0.057 and 0.070 while embedding retrieval over the same encoder reaches 0.73. That is a roughly $10\times$ gap that does not move with K (Table 15). Swapping ArcFace R50 for AntelopeV2 R100 trained on 360K identities did not help either.

Table 15. PATH A after 100K-step continual pretraining on the 2180-identity face pool, at 300 registered identities. Recall is pinned near 0.07 regardless of codebook size, even when the gate routes to the correct code about half the time.

Codebook size K	128	256	512	1024
PATH A recall@1	0.057	0.064	0.070	0.070
gate routes to correct code	0.43	0.51	0.42	0.54
embedding retrieval (same encoder)		0.73		

The reason is a vice the codebook cannot escape, and the diagnostics name it exactly. Two pressures pull in opposite directions as K grows. A *small* codebook packs many people into each cell: at K=16, distinct identities collide in the same code 8.7% of the time, and once two people share a code they are indistinguishable forever after. A *large* codebook fixes that, with inter-identity collisions falling to 2.4% at K=128, but it shatters each identity across cells. The rate at which two cross-condition photos of the *same* person land in the same code drops from 0.33 to 0.20, so the query no longer routes to where the registration was stored. Net recall is squeezed from both sides and never escapes $\sim$0.07. Even when the gate does route a query to the right code (about half the time, Table 15), every other identity sharing that cell is an equally good answer, so the final pick is near-random.

This is the same information loss as captioning, learned instead of written: any categorical bottleneck, whether a word or a code, throws away the continuous signal the encoder worked to preserve, and no amount of compute downstream can recover it. Continuous attention over the raw embedding never quantizes, and never pays this tax. PATH A is in the paper because its failure is the cleanest possible argument for the design that replaced it.

References

Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. MyVLM: Personalizing VLMs for user-specific queries. In *European Conference on Computer Vision (ECCV)*, 2024.

Ruichuan An, Sihan Yang, Ming Lu, Kai Zeng, Yulin Luo, Ying Chen, Jiajun Yong, Huanqia Liang, Qi Huang, and Shanghang Zhang. MC-LLaVA: Multi-concept personalized vision-language model. *arXiv preprint arXiv:2411.11706*, 2024.

Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems*, 2020.

Huiyu Bai, Runze Wang, Zhuoyun Du, Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng, and Xuejiao Zhao. Online-PVLM: Advancing personalized VLMs with online concept learning. *arXiv preprint arXiv:2511.20056*, 2025.

Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *INTERSPEECH*, 2020.

Yuan Gong, Yu-An Chung, and James Glass. AST: Audio spectrogram transformer. In *INTERSPEECH*, 2021.

Haoran Hao, Jiaming Han, Changsheng Li, Yu-Feng Li, and Xiangyu Yue. RAP: Retrieval-augmented personalization for multimodal large language models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

Gary B Huang, Marwan Mattar, Tamara Berg, and Eric Learned-Miller. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. In *Workshop on Faces in Real-Life Images*, 2008.

Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*, 2020.

Bojie Li. Models take notes at prefill: KV cache can be editable and composable. *arXiv preprint arXiv:2606.17107*, 2026a.

Bojie Li. User as code: Executable memory for personalized agents. *arXiv preprint arXiv:2606.16707*, 2026b.

Bojie Li. User as engram: Internalizing per-user memory as local parametric edits. *arXiv preprint arXiv:2606.19172*, 2026c.

Bojie Li and Noah Shi. The latent bridge: A continuous slow–fast channel for real-time game agents. *arXiv preprint arXiv:2606.24470*, 2026.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning (ICML)*, 2023.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Steven R Livingstone and Frank A Russo. The Ryerson audio-visual database of emotional speech and song (RAVDESS). volume 13, 2018.

Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory (M3-Agent). *arXiv preprint arXiv:2508.09736*, 2025.

Stylianos Moschoglou, Athanasios Papaioannou, Christos Sagonas, Jiankang Deng, Irene Kotsia, and Stefanos Zafeiriou. AgeDB: The first manually collected, in-the-wild age database. In *CVPR Workshops*, 2017.

Thao Nguyen, Haotian Liu, Yuheng Li, Mu Yi, Jiarui Mu, and Yong Jae Lee. Yo'LLaVA: Your personalized language and vision assistant. *arXiv preprint arXiv:2406.09400*, 2024.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. LibriSpeech: An ASR corpus based on public domain audio books. 2015.

Karol J Piczak. ESC: Dataset for environmental sound classification. In *ACM Multimedia*, 2015.

Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice. Accessed 2026-06-28.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-Networks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

Letta Team. Letta (formerly MemGPT): Stateful LLM agents with long-term memory. *arXiv preprint arXiv:2310.08560*, 2024. Software framework.

Yu Wang, Dmitry Krotov, Yuanzhe Hu, Yifan Gao, Wangchunshu Zhou, Julian McAuley, Dan Gutfreund, Rogerio Feris, and Zexue He. MemoryLLM: Towards self-updatable large language models. *arXiv preprint arXiv:2402.04624*, 2024.

WikiArt. WikiArt: Visual art encyclopedia. 2010. https://www.wikiart.org.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024.

Yuhuai Wu, Markus N Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations (ICLR)*, 2022.